HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India

Sep 2021 – Jan 2025

- Designed and manufactured prototype units and tooling for new **pressure safety relief devices**, liaising with sub-contract vendors to ensure timely and within-budget component delivery.
- Carried out detailed design of product and necessary tooling utilizing **AutoCAD** (2D/3D) and conducted preliminary **Finite Element analysis** on complex geometries.
- Managed project timelines and resource allocation for the development of new safety products, preparing **periodic status reports** and monitoring project cost consumption.
- Performed validation testing of new designs against project objectives, building and testing prototypes to confirm performance, and maintaining detailed electronic lab notebook records.
- Contributed to the preparation of compliance packages to international standards and oversaw necessary code compliance for new product launches.
- Collaborated with **machine shop equipment** operators and 3D printing technicians to manufacture tooling and prototype parts, accelerating development cycles.
- Supported the transition of new product designs into production, preparing manufacturing and quality documentation and providing training to production and engineering teams.

SKILLS

Engineering Design – Product Design, Tooling Design, Prototype Manufacture, FEA, AutoCAD, Solid Modelling, Risk Assessment, Cost Estimation, Material Processing (Metals & Plastics)

Project Management – Microsoft Project, Planning, Scheduling, Resource Allocation, Budget Monitoring, Status Reporting, Design Reviews, Change Management

Testing & Validation – Performance Testing, Validation Procedures, Lab Notebooks, Electronic Records, Quality Assurance, Technical Code Conformance

Manufacturing & Prototyping – 3D Printing, Machine Shop Equipment, Sub-contract Vendor Management, Budget Management, On-Time Delivery

Software & Tools – AutoCAD R13/R14, Ansys FEA, Microsoft Project, Win 365, Excel, Word, Access, Electronic Instrumentation

Documentation & Compliance – Manufacturing Documentation, Quality Documentation, Compliance Packages, International Standards, BS&B; Company Procedures

PROJECTS

Rupture Disk Design & Validation (HCL Technologies)

- Designed a novel rupture disk assembly for high-pressure industrial applications, focusing on improved burst accuracy and extended service life.
- Manufactured functional prototypes using CNC machining and 3D printing, followed by rigorous performance testing to validate burst pressure and fatigue resistance against design specifications.
- Documented the complete design process, including CAD models, FEA simulations, test results, and prepared a preliminary manufacturing Bill of Materials.

Explosion Vent Prototype Refinement (HCL Technologies)

- Refined the design of an explosion vent product line by modifying panel geometry and sealing mechanisms to enhance ease of installation and reduce manufacturing costs.
- Coordinated with external machine shops for component fabrication and assembled multiple prototype units for functional and environmental stress testing.
- Analyzed test data to confirm improved performance characteristics and contributed to the creation of initial production support documentation.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)